

HINTS AND SOLUTIONS

- Q.1 D
Q.2 D
Q.3 B
Q.4 D
Q.5 B
Q.6 A
Q.7 D
Q.8 D
Q.9 B
Q.10 C
Q.11 B
Q.12 B
Q.13 A

Sol. $x = A \sin x (\omega t)$
 $5 = A \sin (\omega t)$
 $v = A\omega \cos \omega t$
 $5 = 10 \cos (\omega t)$

$$\cos \omega t = \frac{1}{2} \Rightarrow \omega t = \frac{\pi}{3}$$

$$5 = A \times \frac{\sqrt{3}}{2}$$

$$A = \frac{10}{\sqrt{3}} \Rightarrow \omega = \frac{10}{\frac{10}{\sqrt{3}}} = \sqrt{3} \text{ rad/s}$$

$$t = \frac{\pi}{3\omega} = \frac{\pi}{3\sqrt{3}} \text{ sec.}$$

- Q.14 A
Q.15 A

Q.16 **Ans :** $\left\{ \frac{(\pi\sqrt{5})}{6\sqrt{3}} \right\} \text{ s}$

Q.17 **Ans:** $\pi(c - b) / V$.

Q.18 **[Ans : 4 J]**

Q.19 **Ans :** $10 \sin (\pi t + \pi / 6) \text{ cm}$

Q.20 **[Ans :** $2\pi t_0 / \cos^{-1} \left(\frac{a+c}{2b} \right)]$

Q.21 **[Ans :** $2\pi \sqrt{\frac{m(kl^3 + 3YI)}{3YIk}}]$

Q.22 $\left[\text{Ans : (a)} A \left[\sin^2(\omega t_1) + \frac{M}{M+m} \cos^2(\omega t_1) \right]^{1/2} \text{ (b) } \omega \sqrt{\frac{M}{M+m}} \right]$

Q.23 $\left[\text{Ans : } \frac{1}{2\pi} \sqrt{\frac{mgL + kh^2}{mL^2}} \right]$

Q.24 $[\text{Ans : } 2\pi \sqrt{\frac{3m}{2k}}]$

Q.25 $[\text{Ans : } 999, 1000, 1001 \text{ rad/s}]$

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